# Supplementary Material

## Updating the World Is Not Rebinding the Target

Anonymous Submission

**Review note.** This document supports the main submission with implementation, protocol, and extended-result details. The main paper is self-contained. No result reported here changes the primary estimand, and all files are included in the accompanying anonymous artifact rather than referenced through an external webpage.

# 1 Benchmark and Frozen Protocol

The TRI-v3 primary evaluation contains 160 tasks in 20 frozen language-template clusters. Tasks are balanced across anchored and dynamic references and across flip, stable, remove, invalidate, and name-collision transitions. The transfer evaluation contains 80 tasks over four unseen schemas (projects, expenses, inventory, and cloud deployments). The guarded TRI-v4 extension contains 40 conditional tasks and distinguishes action-validity and selector-match guards. The model-facing SQLite replay uses a frozen 40-task subset and records every query, refresh, target proposal, mutation attempt, and final database diff.

The task generator evaluates the selector in $S_0$, applies a controlled transition to obtain $S_1$, and computes the gold target from the instruction's reference mode and policy. Paired items share their schema, entities, transition, and action; only the temporal discourse varies. Gold targets and benchmark metadata are not passed to the model.

# 2 Controller Records

The Generic pipeline passes the original instruction separately to its actor; its compiled ledger stores the initial selected ID, full entity snapshot, selector, requested action, and action preconditions. It omits reference mode and invalidity/fallback semantics. The scalar Lifecycle compiler additionally emits `reference_mode`, `bound_target_id`, `selector`, and `invalidity_policy`. A deterministic executor checks action validity for preserved identities and delegates only explicitly dynamic or unresolved branches to the actor. The separate TRI-v4 conditional extension adds `guard_type` and `fallback_policy`; these fields are not part of the scalar v3 record. The exact historical Compile-then-act baseline is retained unchanged as a neighboring audit.

## 2.1 Lifecycle-Gated procedure

| | Lifecycle-Gated execution |
|---|---|
| 1 | Observe $S_0$ and compile $L(r, a)$ from the instruction and action schema. |
| 2 | Execute the required environment refresh and receive $S_1$. |
| 3 | If mode is PRESERVE, check the compiled bound ID against $V_a$. |
| 4 | If valid, emit the bound ID without invoking the actor. |
| 5 | If invalid, apply the compiled reject or reselection fallback. |
| 6 | For REEVALUATE or unresolved conditional branches, invoke the actor on $L(r)$ and $S_1$. |

Table 1: Controller procedure. Gold targets and benchmark selector fields are never provided to the compiler or actor.

## 2.2 Information and execution audit

| Controller | Pre-refresh output | Calls | Gate |
|---|---|---|---|
| Generic ledger | ID, snapshot, selector, action, preconditions | 2 | no |
| Untyped plan | free-form plan | 2 | no |
| Historical CTA | binding time, ID, selector, reason | 2 | no |
| Lifecycle free | mode, ID, selector, policy | 2 | no |
| Lifecycle gated | same typed record | 1 or 2 | yes |

Table 2: The gated Preserve branch omits the actor call. Generic retains all task-relevant evidence and is not an information-deletion baseline.

Untyped-plan and Lifecycle-free both compile before refresh, use two calls, share the action schema and actor, and differ in whether the intermediate contract is free-form or typed. Historical CTA is not a pure representation match: its actor omits the action schema and its Preserve rule rejects only missing, rather than present-but-invalid, entities.

## 2.3 Executor-level oracle check

| Generator-supplied representation | Accuracy | Characteristic failure |
|---|---|---|
| Bound ID only | 41.9 | premature binding |
| Binding time only | 58.1 | no anchored identity |
| Bound name only | 58.9 | alias/name collision |
| Latest state | 64.6 | temporal rebinding |
| Time + ID | 74.0 | invalidity/conditional policy |
| Full lifecycle schema | 100.0 | – |

Table 3: Deterministic execution over 246 generated tasks. This establishes sufficiency for generated semantics only, not model performance, minimality, or natural-language validity.

# 3 Extended Component Audit

| Controller | Qwen3.5 | GLM-5.1 |
|---|---|---|
| Generic free actor | 64.4 | 71.9 |
| Generic + reference mode | 75.0 | 75.0 |
| Generic + validity gate | 65.0 | 73.1 |
| Untyped pre-refresh plan | 81.2 | 70.6 |
| Exact Compile-then-act | 95.0 | 96.2 |
| Lifecycle free actor | 96.9 | 98.1 |
| Lifecycle + gate | 98.1 | 100.0 |

Table 4: Exact target accuracy on the identical 160-task language set.

The post-freeze Generic+reference-mode ablation adds only the explicit mode field and reaches 75.0% for both models. Its paired cluster gains over Generic are +10.6 $[+5.0, +16.9]$ for Qwen and +3.1 $[-2.5, +9.4]$ for GLM; the latter includes zero. The ablation therefore supports explicit mode classification as a component, but not as a complete explanation of CTA's gain. The audit separates three effects. A validity-only gate changes Generic accuracy by only 0.6 and 1.2 points. Lifecycle state with a free actor reaches 96.9 and 98.1%, while the deterministic gate adds 1.2 and 1.9 points. Exact Compile-then-act reaches 95.0 and 96.2%. The call-matched untyped plan reaches 81.2 and 70.6%; Lifecycle-free exceeds it by 15.6 and 27.5 points with

| Policy control | Overall | Anchored | Dynamic |
|---|---|---|---|
| Always-Lock + validity | 96/160 (60.0) | 80/80 (100.0) | 16/80 (20.0) |
| Always-Reevaluate | 96/160 (60.0) | 16/80 (20.0) | 80/80 (100.0) |

Table 5: Deterministic policy extremes on the same frozen inventory. These controls are not LLM runs or reproductions of a named external agent; they expose the complementary errors of unconditional locking and unconditional selector re-evaluation. The machine-readable report is `reports/policy_extreme_controls.md`.

cluster intervals excluding zero. The evidence therefore supports structured pre-refresh semantic compilation as the main accuracy mechanism, with typed lifecycle state providing an auditable interface and deterministic enforcement providing a smaller safety contribution.

## 3.1 Referential-core versus reject-policy sensitivity

The pre-specified aggregate includes 32 anchored remove/invalidate items whose gold execution outcome is the author-specified `INVALID_BOUND_ENTITY`. Table 6 separates these from the 128 actionable referential-core items without redefining benchmark gold or replacing the aggregate estimand. On the actionable core,

| Run | Actionable core | Reject policy |
|---|---|---|
| Qwen Generic | 95/128 (74.2) | 8/32 (25.0) |
| Qwen Generic + validity gate | 95/128 (74.2) | 9/32 (28.1) |
| Qwen Untyped plan | 106/128 (82.8) | 24/32 (75.0) |
| Qwen Historical CTA | 126/128 (98.4) | 26/32 (81.2) |
| Qwen Lifecycle-free | 123/128 (96.1) | 32/32 (100.0) |
| Qwen Lifecycle-Gated | 125/128 (97.7) | 32/32 (100.0) |
| GLM Generic | 93/128 (72.7) | 22/32 (68.8) |
| GLM Generic + validity gate | 93/128 (72.7) | 24/32 (75.0) |
| GLM Untyped plan | 107/128 (83.6) | 6/32 (18.8) |
| GLM Historical CTA | 127/128 (99.2) | 27/32 (84.4) |
| GLM Lifecycle-free | 125/128 (97.7) | 32/32 (100.0) |
| GLM Lifecycle-Gated | 128/128 (100.0) | 32/32 (100.0) |

Table 6: Correct/total and accuracy in parentheses. The policy slice measures compliance with the benchmark's explicit execution rule, not equally human-supported referential semantics.

Gated-minus-Generic differences are +23.4 points with a template-cluster interval of [+11.3,+38.4] for Qwen and +27.3 [+16.9,+39.8] for GLM. Gated-minus-Historical-CTA is $-0.8$ $[-4.9, +3.1]$ and +0.8 [0.0,+2.6], respectively; the cross-model evidence therefore does not establish Gated superiority over the strong pre-refresh predecessor. Complete reject-policy contrasts are in the artifact report.

## 3.2 Frozen error taxonomy

| Error label | Operational meaning |
|---|---|
| Temporal rebinding | A committed pre-refresh target is replaced by the refreshed selector winner. |
| Premature binding | A dynamic reference incorrectly retains a pre-refresh entity. |
| Invalid processed | A preserved entity violates action preconditions but is still selected. |
| Unneeded rejection | The controller returns $\perp$ although a valid gold target exists. |
| Selector grounding | Mode or guard is correct, but the compiler or actor misapplies a filtered selector. |

Table 7: Mechanism-oriented labels used in stage and write-consequence reports.

# 4 Statistical and Audit Details

The primary comparison is Lifecycle-Gated minus Generic Structured Ledger on Qwen's 20 language clusters. We resample complete clusters 10,000 times with seed 20260717. The primary Qwen difference is +33.8 points with a cluster interval of [+18.1,+50.0]; the GLM replication difference is +28.1 points with [+18.1,+38.1]. Exact paired McNemar tests are secondary: Qwen has 3 Generic-only versus 57 Lifecycle-only wins, $p = 6.25 \times 10^{-14}$; GLM has 0 versus 45, $p = 5.68 \times 10^{-14}$. API errors and missing rows are counted as failures in the audit, although the reported v3/v4 complete runs contain zero API errors and zero retries.

## 4.1 TRI-v7 Core Replication

The post-primary replication was frozen before its model calls with SHA-256 prefix `2504f4979f1b`; the artifact manifest records the complete hash. It contains 240 tasks over ten schemas unused in v3 primary or transfer, four independently parameterized state instances per schema, and balanced binding, update, and explicitness factors. All old targets remain present and action-valid. We resample the 40 state-instance clusters.

| Model | Controller | E2E target | Initial bind | Cond. TRI | Wrong writes | Stable err. |
|---|---|---|---|---|---|---|
| Qwen3.5 | Generic | 47.5 | 107/120 | 43/72 | 44/240 | 4/40 |
| Qwen3.5 | Exact CTA | 70.8 | 103/120 | 0/71 | 8/240 | 6/40 |
| Qwen3.5 | Gated | 71.2 | 95/120 | 0/64 | 17/240 | 9/40 |
| GLM-5.1 | Generic | 70.0 | 120/120 | 38/80 | 38/240 | 2/40 |
| GLM-5.1 | Exact CTA | 94.2 | 104/120 | 0/70 | 14/240 | 0/40 |
| GLM-5.1 | Gated | 97.1 | 119/120 | 0/79 | 7/240 | 0/40 |
| DeepSeek-V4-Pro | Generic | 73.8 | 119/120 | 59/79 | 60/240 | 0/40 |
| DeepSeek-V4-Pro | Exact CTA | 91.2 | 107/120 | 0/70 | 17/240 | 1/40 |

Table 8: Frozen new-state replication with distinct denominators. E2E target uses all 240 rows; initial binding uses 120 anchored rows; conditional TRI requires a correct binding, a changed winner, and a still-valid old entity; wrong writes count all non-gold mutations; stable error uses 40 anchored Stable controls.

Generic conditional drift is 59.7% for Qwen, state-cluster 95% CI [46.5,72.4], and 47.5% for GLM [35.0,61.3]. CTA-minus-Generic overall differences are +23.3 points [16.2,30.4] and +24.2 [19.6,28.7]. SQLite replay turns all 43 and 38 Generic core drifts into wrong-entity writes. CTA/Gated produce zero core TRI writes after a correct binding, but still produce 8/17 Qwen and 14/7 GLM wrong writes from opposite-mode or selector errors. Thus the replication supports the conditional phenomenon and commitment compilation, not general write safety. The original six Qwen/GLM 240-row runs contain zero API, parse, or internal errors. GLM CTA records one successful transport retry; the other five runs record none.

The later DeepSeek run uses the same frozen 240 tasks and inference settings. Generic drifts on 59/79 opportunities, 74.7% with state-cluster interval [62.5,86.1], while CTA drifts on 0/70. CTA-minus-Generic accuracy is +17.5 points [10.8,23.3]. All 59 Generic core drifts become wrong-entity SQLite writes. Across all errors, Generic and CTA produce 60 and 17 wrong writes; CTA's remaining errors are outside the conditional TRI mechanism. Both 240-row files have zero API/parse errors and zero retries.

## 4.2 Matched Full-History Baselines

After the v7 and DeepSeek results were known, we froze a separate strong-baseline protocol on the unchanged 240 tasks. `interactive` is an ordinary two-call trajectory: the first response requests refresh and the second retains the complete conversation, without TRI or reference-mode terminology. `full_history_once` is a one-call semantic upper baseline that sees both states and is explicitly asked whether to preserve, reevaluate, or reject. Exact CTA uses the existing matched v7 run. Temperature is zero, thinking is disabled, and the output cap is 1,200.

All six files contain 240 rows and zero final API/parse failures. Qwen aware uses 269 request attempts with 29 retries; the other one-call aware runs use 240 attempts and zero retries. Interactive runs use 480 attempts. Since neither full-history condition exposes a separately scored pre-refresh selection, its substitution rate is

| Model | Interactive | Aware | CTA | Int. subst. | Aware subst. | CTA–Aware CI |
|---|---|---|---|---|---|---|
| Qwen3.5 | 63.3 | 69.6 | 70.8 | 77/80 | 56/80 | +1.2 [-6.7,9.2] |
| GLM-5.1 | 67.1 | 80.8 | 94.2 | 74/80 | 38/80 | +13.3 [8.8,17.9] |
| DeepSeek | 68.8 | 75.8 | 91.2 | 68/80 | 42/80 | +15.4 [9.2,21.7] |

Table 9: Matched v7 exact-target accuracy (%) and anchored changed-winner substitutions. Intervals resample 40 state clusters. CTA ties the aware Qwen baseline in overall accuracy.

unconditional and is not conditional TRI. SQLite replay converts the interactive errors into 87/79/75 wrong writes and the aware errors into 70/46/57 wrong writes for Qwen/GLM/DeepSeek, with four invalid attempts and one unnecessary rejection across all six runs. Thus complete history and a late semantic reminder do not remove wrong-entity consequences; the result does not establish universal overall CTA superiority.

The Qwen 40-task SQLite run gives Generic Structured Ledger 27/40 final states and 13 wrong-entity writes; the GLM run gives 26/40 and 8 wrong-entity writes. A conditional audit requires a correct initial binding, an anchored flip/name-collision update, continued presence and action-validity of the old entity, and a final write to the refreshed winner. Under this strict definition, core TRI writes occur in 8/8 Qwen and 6/8 GLM opportunities, versus 0/4 stable controls for each model. The remaining 5 and 2 wrong writes follow removal or invalidation and are reported separately as reject-policy errors. On Qwen, Generic plus a validity-only gate remains 27/40, while Lifecycle-free and Lifecycle-Gated reach 40/40. The separately run GLM Lifecycle-Gated replay also reaches 40/40; GLM Lifecycle-free was not run on SQLite. All reported 40/40 cells have zero wrong writes and collateral modifications. These are controlled mutation results, not a claim about arbitrary unconstrained agents: they assume the lifecycle record was compiled correctly.

# 5 External Boundary Check

The ToolSandbox-based 24-task pilot is exploratory and is not an official benchmark score. For Generic, Lifecycle, Gated, and Untyped, respectively, accuracies are 79.2/87.5/91.7/91.7% for GLM and 91.7/83.3/91.7/79.2% for Qwen. Gated includes one GLM and two Qwen wrong writes. It therefore demonstrates that the target-integrity diagnosis can be instantiated in a stateful external-style tool loop, but does not establish a stable method advantage outside the controlled TRI setting.

A post-hoc strict audit joins each trace to the frozen manifest and counts only Preserve/Flip rows where the compiled initial ID is correct, the old entity remains present, and the refreshed winner is a different stable ID. GLM Generic writes the refreshed winner in 3/6 opportunities, with 0/2 wrong writes on eligible Stable controls; GLM Lifecycle is 0/5. Qwen Generic is 0/6, while Qwen Lifecycle-free is 2/6 and deterministic gate replay reduces this to 0/6. Untyped plans expose no auditable compiled ID and are excluded from this conditional denominator. The violating GLM tasks span newest-created, oldest-created, and alphabetically-last selectors. This is positive evidence that TRI can occur in a benchmark-compatible database/tool substrate, but the inventory has only six selector clusters and the strict audit was specified after the outputs existed. We therefore report task IDs and no confirmatory interval or prevalence claim.

We then ran a separate frozen 96-task single-user-turn ToolSandbox-based extension with a Preserve/Reevaluate × Stable/Flip design and an auditable initial-binding opportunity. Qwen and GLM full-history agents had zero conditional TRI mechanism errors in 70 and 73 opportunities; matched Qwen and GLM Generic Ledgers had zero in 64 and 87 opportunities. The runs still produced 6, 13, 5, and 4 wrong-entity writes, respectively, but all were initial selector or tool ordering errors and therefore excluded from the post-binding numerator. This negative external result is retained as a boundary check rather than merged with the 24-task pilot or interpreted as a prevalence estimate.

## 5.1 Official opportunity and released-trace audit

We separately audited the unmodified public task inventories rather than multiplying semantic tasks by tool-description augmentations. At ToolSandbox commit `165848b`, 129 semantic scenario families generate 1,032 tool-presentation instances. None contains the strict sequence correct binding, independent external transition, and later substitutable entity mutation. One multiple-user-turn task is TRI-like: after finding all friends and changing those stable person IDs to enemies, "them" refers to the same IDs when changing them back. It has no external refresh, competitor, or selector flip.

The downloaded AppWorld `0.1.3.post1` release contains 732 task instances from 244 generator families. It likewise has no independently scheduled post-binding transition, but Todoist family `8ce6779` supplies a natural action-induced near-match. The Agent selects incomplete tasks assigned to the user, reassigns them, and then must "leave a comment there" on the same task IDs even though reassignment makes them fail the initial selector. We audited all 42 released trajectories for this family across 14 experiment configurations. Among 16 operations on gold task IDs, all 16 subsequent comments preserve the same ID and none substitutes another target. There are 152 omitted gold targets, two assignments to non-gold IDs, and three comments on non-gold IDs; only two trajectories pass every official test. These are upstream selection and coverage failures, not post-binding substitutions after a correct target operation.

At current $\tau^3$-bench commit `cf71a80`, we further audit 2,449 core tasks: 50 airline, 114 retail, and 2,285 telecom. Airline and retail expose no user-side mutation during conversation. Telecom has 2,250 tasks with user mutation actions, but none carries a customer, line, or bill stable ID, and its single-device/app actions create no competing same-role selector target. The strict native TRI count is therefore zero. Eight telecom tasks are natural dual-control near-matches: the Agent binds an overdue bill, the user pays it, and the Agent resumes an associated line. Bill and line are different referential roles, so these are not TRI. The 26 released result files contain 10,832 trajectories from GPT-4.1, GPT-4.1-mini, Claude 3.7, and o4-mini; 936 trajectory IDs match the payment family, but none supplies a strict TRI denominator.

### 5.2 Custom AppWorld database case study

The custom AppWorld study uses two application/selector clusters. The Todoist cluster was frozen first and uses native `create_task`, `show_tasks`, and `update_task` APIs. A post-primary Simple Note extension uses native `create_note`, `search_notes`, and `add_content_to_note`. Each cluster crosses Preserve/Reevaluate, Stable/Flip, and two paraphrases. Todoist selects the earliest-due incomplete task; Stable adds a later task and Flip an earlier one. Simple Note selects the alphabetically first tagged title; Stable adds `DELTA` and Flip adds `ALPHA`. Old targets remain present and editable. Gold is computed from stable IDs, and evaluators compare post-sync and final database snapshots without an LLM judge.

| App | Agent | Strict | Write | Bind | TRI | Wrong |
|---|---|---|---|---|---|---|
| Todoist | Qwen3.5 | 6/8 | 8/8 | 6/8 | 0/6 | 0 |
| Todoist | GLM-5.1 | 8/8 | 8/8 | 8/8 | 0/8 | 0 |
| Simple Note | Qwen3.5 | 6/8 | 7/8 | 6/8 | 0/6 | 1 |
| Simple Note | GLM-5.1 | 4/8 | 8/8 | 4/8 | 0/4 | 0 |

Table 10: Custom AppWorld boundary study. Across 24 correct, correctly timed bindings, conditional TRI is 0/24. The one Simple Note wrong write follows synchronization before binding and is excluded from that denominator. Explicit sidecar omission causes the remaining strict failures. This is not an AppWorld leaderboard result; each app contributes one selector cluster.

A post-primary lower-intervention addendum removes `record_binding` and all prompt language about TRI, binding, or temporal authorization. Each ordinary selector API returns one stable ID, which is logged automatically as the binding event. Across 32 Qwen/GLM trajectories, strict success is 28/32, authorized writes are 30/32, and conditional TRI is 0/28 (0/6 in Preserve/Flip). Qwen's two wrong writes use the same Preserve wording in both apps: it synchronizes before its first selector call, then writes the refreshed winner. Matched Stable rows exhibit the same order defect but happen to write A because the winner is unchanged. This is a reproducible pre-binding temporal-order failure and illustrates why Stable controls and a conditional TRI denominator are both necessary.

## 6 Negative Compositional Stress Tests

The frozen Qwen TRI-v5 stress test contains 40 SQLite tasks with two refreshes, a monitoring referent, an unrelated tool call, and a final mutation. Generic obtains 32/40 final states, one wrong write, and four unnecessary rejections. Scalar Lifecycle obtains 28/40, six wrong writes, six unnecessary rejections, and one parse/internal error. The difference is $-10.0$ points with a template-cluster interval of $[-35.0, 12.5]$. Lifecycle mode accuracy is only 25/40: its scalar record confuses monitoring and action-target roles.

A frozen role-indexed held-out addendum then tests 40 tasks from four unseen schemas. In the matched scalar-versus-role comparison, Qwen improves from 35/40 to 39/40 final states (difference +10.0, cluster CI [+2.5,+20.0], McNemar $p = .21875$). GLM intention-to-treat is 40/40 scalar versus 37/40 role-indexed because three role runs fail in transport; a serial recovery sensitivity gives 40/40 for both. Against historical CTA, Qwen obtains 38/40 versus 39/40 final states (33/40 vs. 39/40 target accuracy), and recovered GLM ties at 40/40. These results motivate role indexing but do not support stable superiority over simpler compilation.

# 7 Post-Primary Method-Upgrade Go/No-Go

After the scalar and compositional results above were known, we froze a 20-task decision set: 16 v7 core tasks over 16 state clusters and four v6 multi-refresh/role tasks. We reused the matching frozen Generic, Exact CTA, Lifecycle, and role-indexed outputs and newly ran an Event Graph compiler (M1) and Event Graph plus Executable Selector (M2) with both models. This is an exploratory method-selection smoke, not a powered confirmatory comparison.

| Model | Method | Correct | Schema | Mode | Selector |
|---|---|---|---|---|---|
| Qwen3.5 | Event Graph | 9/20 | 14/20 | 12/20 | – |
| Qwen3.5 | Executable Selector | 15/20 | 15/20 | 14/20 | 15/20 |
| GLM-5.1 | Event Graph | 20/20 | 20/20 | 20/20 | – |
| GLM-5.1 | Executable Selector | 18/20 | 18/20 | 18/20 | 19/20 |

Table 11: Post-primary 20-task method-upgrade smoke. Selector reports equivalence on both initial and final states. Parse/schema failures count as incorrect under intention-to-treat.

On the same combined tasks, Exact CTA obtains 13/20 for Qwen and 20/20 for GLM. M2 therefore changes accuracy by +2 and −2 tasks, respectively, and fails the predeclared requirements that both models reach at least 95% schema/selector equivalence and show a consistent direction. We do not promote M1 or M2 to the main method. Exact CTA remains the scalar mechanism, while the independently frozen v6 evidence above motivates role indexing as a limited compositional extension. The zero-API oracle executor is 440/440 on v3/v7/v6 and blocks all stale/wrong writes in 120 deterministic concurrency sequences; this establishes implementation sufficiency only, not learned compiler performance.

**Binding Drift audit.** The two projects can be located on two independent decision axes:

| Initial binding | Preserve after update | Reevaluate after update |
|---|---|---|
| Correct | TRI-Preserve | TRI-Reevaluate |
| Incorrect | Error propagation/repair | Repair plus authorized reselection |

Table 12: Problem map. Binding Drift primarily studies initial binding correctness and its propagation; TRI conditions on a correct binding and varies whether later reevaluation is authorized. The lower-right combination is outside the present TRI claim.

We audited the official Binding Drift repository at commit `0e040e0954b1`. Its documented offline command references two files deleted at that commit; exposing the current `workflows_200.jsonl` under the legacy filename in a temporary directory yields 25/25 assertions with unmodified metric and defense code. Its deterministic re-verification reads a gold target and is an oracle upper bound, whereas Entity Lock corresponds to our Always-Lock extreme. We therefore do not place its oracle score in the learned-method main table. Our matched one-shot aware baseline sees the original instruction and both state epochs and must resolve whether to preserve, reevaluate, or reject; it is a practical post-refresh re-resolution approximation, not the official self/cross method. The official learned re-verifiers have not been adapted to expose a separately scored pre-refresh binding or a user-authorized deferred query, so we do not label that comparison as an official reproduction.

After freezing a 20-task symmetric smoke (ten v7 domains, one identical-state Preserve/Reevaluate Flip pair per domain), we ran an explicit author adaptation. The official re-verifier normally receives a short `step1_referent`; using only TRI's selector would delete the temporal authorization variable, so the official prompt frame receives the complete TRI instruction and refreshed candidates. Gold, mode labels, and CTA state are hidden.

| Model | Policy | Overall | Preserve | Reevaluate |
|---|---|---|---|---|
| Qwen | Lock analogue | 10/20 | 10/10 | 0/10 |
| Qwen | Self reverify | 3/20 | 0/10 | 3/10 |
| Qwen | Cross (GLM) | 10/20 | 0/10 | 10/10 |
| Qwen | Frozen CTA | 12/20 | 8/10 | 4/10 |
| GLM | Lock analogue | 10/20 | 10/10 | 0/10 |
| GLM | Self reverify | 10/20 | 0/10 | 10/10 |
| GLM | Cross (Qwen) | 3/20 | 0/10 | 3/10 |
| GLM | Frozen CTA | 17/20 | 7/10 | 10/10 |

Table 13: Post-primary Binding Drift author-adaptation smoke. Cross rows reuse the other model's independent verifier output on the same input, avoiding duplicate calls. This is not an official Binding Drift score. Qwen reverify selects another visible but selector-ineligible entity on 14/20 tasks; GLM selects the refreshed winner on all 20.

Both verifier runs contain 20/20 expected rows, zero API/parse errors, and zero retries (16,088 tokens total). Thus GLM cleanly exposes complementary lock/re-resolution errors, while Qwen mainly exposes an interface grounding failure. CTA is not perfect, but is the only tested policy with nonzero accuracy on both modes for both models. These results do not test initial-misbinding repair and cannot establish superiority on Binding Drift's original benchmark.

**Leave-group-out sensitivity.** On the 240-task v7 replication, we recompute the matched CTA-minus-Generic difference after excluding one complete domain or one complete template family. The ten domain exclusions retain ranges of 18.5–25.5, 23.1–25.9, and 15.7–22.2 points for Qwen, GLM, and DeepSeek; the sixteen template-family exclusions retain 20.4–26.3, 20.8–26.1, and 14.2–19.6 points. Every exclusion remains positive. This rules out dominance by one domain or template family, but does not make templated rows independent natural-world observations.

**Retrospective sample-sufficiency audit.** We additionally subsample complete matched clusters without replacement 10,000 times at increasing cluster counts. This is a precision and stability diagnostic, not post-hoc power. On v3, the full Lifecycle-Gated–Generic differences are +33.8 points [18.7,49.4] for Qwen and +28.1 [18.1,38.1] for GLM under template-cluster bootstrap. At only 10 of 20 template clusters (80 tasks), 100% of subsamples retain a positive difference for both models. On v7, the full CTA–Generic differences are +23.3 [16.2,30.4], +24.2 [19.6,28.7], and +17.5 [10.8,23.3] for Qwen, GLM, and DeepSeek. At 10 of 40 state clusters (60 tasks), positive differences occur in 100.0%, 100.0%, and 99.8% of draws; at 20 clusters all three reach 100%. Thus the controlled paired effect is not sample-limited within the observed inventory. This does not increase the number of natural-world opportunities or justify a prevalence claim.

**Temperature-zero repeat stability.** Before repeat calls, we froze one balanced task from each of the 40 v7 state clusters. The matching subset of the original run is repeat 1; two new complete passes give three measurements for Qwen and GLM Generic and CTA. CTA–Generic remains positive in every model-repeat cell: Qwen differences are +25.0, +12.5, and +22.5 points, while GLM differences are +30.0, +20.0, and +25.0. Generic conditional drift is 5/10, 5/11, and 7/11 for Qwen and 4/12, 5/12, and 6/12 for GLM; CTA is zero in every denominator. Exact target unanimity across all three passes is 70.0/72.5% for Qwen Generic/CTA and 85.0/97.5% for GLM. Under the frozen rule this is a Mixed result: the method direction and conditional mechanism are stable, but individual outputs are not fully deterministic. The 320 new task-controller executions use 640 requests, zero retries, and 313,692 provider-reported tokens, with zero API/parse errors.

# 8 Request Cost

On the 160-task primary set, Generic uses 320 requests per model. Lifecycle-Gated uses 80 Preserve compiler calls and 160 Reevaluate compiler/actor calls, totaling 240. Mean client-observed Qwen latency is 2.84 seconds for gated Preserve versus 3.63 for Generic anchored tasks; corresponding GLM means are 3.47 versus 5.18 seconds. On the 40-task SQLite set, Generic uses 80 requests and Gated uses 60. Token usage was not captured and latency is network-dependent, so these are request-efficiency observations rather than a controlled systems benchmark.

For the post-primary DeepSeek v7 run, both Generic and CTA use 480 requests. Provider-reported usage is 264,359 tokens for Generic and 216,446 for CTA; client-observed cumulative latency is 1,651.7 and 1,355.8 seconds. These matched-endpoint observations are reported for provenance, not as a cross-provider systems claim.

# 9 Blind Human Construct Validation

One volunteer uninvolved in TRI design naturally rewrote 50 sampled scalar instructions. Three other volunteers independently labeled 100 differently randomized original/rewrite items. Forms used opaque IDs and concealed source task, binding mode, update, variant, and gold. Responses were an entity ID, Reject, or Clarify; confidence was optional. All three files contain exactly the 100 keyed IDs with no missing, duplicate, invalid, or payload-mismatched rows. No two annotators have identical response vectors.

| Slice | $n$ | Maj.–gold | Unanimous | $\kappa$ | $\alpha$ |
|---|---|---|---|---|---|
| All | 100 | 86.0 | 72.0 | .708 | .709 |
| Original | 50 | 84.0 | 70.0 | .679 | .681 |
| Human rewrite | 50 | 88.0 | 74.0 | .737 | .738 |
| Explicit | 60 | 86.7 | 70.0 | .672 | .674 |
| Implicit | 40 | 85.0 | 75.0 | .758 | .760 |
| Dynamic | 50 | 98.0 | 96.0 | .924 | .925 |
| Anchored actionable | 30 | 86.7 | 63.3 | .538 | .543 |
| Anchored reject | 20 | 55.0 | 25.0 | .087 | .102 |

Table 14: Human agreement (percent except coefficients). Majority–gold uses all items, treating six three-way ties as not matching gold. Among 94 determinate majorities, accuracy is 91.5%.

Pairwise raw agreement is 79.3%; 4% of items receive at least one Clarify. All 72 unanimous items match gold. Annotator-level gold agreement is 87%, 80%, and 89%; confidence was absent for annotator 1 and therefore is not used as an outcome. The rewrite slice does not degrade agreement, and explicit/implicit slices are similar. The main weakness is execution fallback: participants often preserve an invalid entity as the discourse referent while disagreeing whether the action answer should be Reject. We therefore interpret the study as support for the Preserve/Reevaluate core, not as validation of reject or conditional policy. Complete disagreement rows and the auditable analysis script accompany the artifact.

**Human-majority model sensitivity.** Of the 50 original items, 46 have a determinate human majority, 42 majorities match benchmark gold, and 35 are unanimously gold-aligned. Re-scoring frozen model outputs against the human majority gives: Gated-minus-Generic template-cluster gaps are +19.6 points with a 95% interval of [+5.0,+36.4] for Qwen and +17.4 [+6.8,+31.6] for GLM (18 clusters, 10,000 draws). Secondary task-level exact McNemar tests give $p = .0117$ and .0078. On the 42 majority-gold items, Gated reaches 97.6% and 100%; on the 35 unanimous-gold items it reaches 97.1% and 100%. This sensitivity preserves the Generic-to-compiled gap but does not show Lifecycle-Gated superiority over historical pre-refresh compilation.

| Model | Generic | Lifecycle-Gated | Historical CTA |
|---|---|---|---|
| Qwen3.5 | 32/46 (69.6) | 41/46 (89.1) | 42/46 (91.3) |
| GLM-5.1 | 34/46 (73.9) | 42/46 (91.3) | 41/46 (89.1) |

Table 15: Accuracy against determinate human-majority targets (percent in parentheses).

# 10 Independent Human-Rewrite Model Evaluation

After the human study and before model calls, we froze 50 instructions rewritten by one independent volunteer and reran the unchanged four controllers. All eight files contain exactly 50 unique expected IDs, all 400 rows have `status=ok`, and no request retry occurred. On the 37 unanimous-gold rewrites,

| Model | Generic | CTA | Lifecycle-free | Gated |
|---|---|---|---|---|
| Qwen3.5, benchmark gold | 60.0 | 90.0 | 88.0 | 90.0 |
| GLM-5.1, benchmark gold | 74.0 | 98.0 | 92.0 | 94.0 |
| Qwen3.5, human majority | 60.4 | 89.6 | 83.3 | 85.4 |
| GLM-5.1, human majority | 68.8 | 93.8 | 83.3 | 85.4 |
| Qwen3.5, human actionable | 68.3 | 92.7 | 80.5 | 82.9 |
| GLM-5.1, human actionable | 68.3 | 92.7 | 80.5 | 82.9 |

Table 16: Accuracy (%) on all 50 rewrite golds, the 48 determinate rewrite majorities, and the 41 majority-actionable items (excluding majority-Reject).

Generic/CTA/Lifecycle-free/Gated reach 75.7/97.3/86.5/89.2% for Qwen and 81.1/100/89.2/91.9% for GLM. On the 25 dynamic items, CTA/Gated reach 96/80% and 100/88%, respectively; complete explicitness, update, template, error-type, and majority-gold slices are in the artifact. CTA-minus-Generic benchmark-gold differences are +30.0 points with a template-cluster interval of [+11.3,+50.0] for Qwen and +24.0 [+8.3,+41.3] for GLM. Gated reaches the same Qwen gold accuracy as CTA but is four points lower for GLM and lower against human majorities for both models. Direct CTA-minus-Gated intervals are 0.0 $[-11.1, +11.1]$ for Qwen and +4.0 $[-4.3, +14.8]$ for GLM; CTA-minus-Lifecycle-free intervals are +2.0 $[-8.5, +12.2]$ and +6.0 $[-3.6, +16.7]$, respectively. None excludes zero, so the experiment does not establish CTA superiority over either typed variant. All five Qwen and three GLM Gated benchmark-gold errors are dynamic items compiled as Preserve; one is human-contested. CTA's errors are predominantly author-specified invalid-target outcomes. The result therefore supports pre-refresh commitment compilation on this volunteer-authored rewrite set, not Lifecycle-Gated superiority. The execution log discloses two pre-call empty-run failures and the non-balanced health-smoke deviation.

# 11 Artifact Map

The accompanying anonymous artifact contains the frozen JSONL task files, raw model outputs, SQLite traces, report scripts, controller implementations, protocol files, SHA-256 manifests, and tests. The directory `tri_artifact/reports/` contains the primary cluster reports, component audits, SQLite replay reports, ToolSandbox pilot report, and claim-provenance records, official opportunity/trace audits, and the AppWorld custom protocol. Current submission runners receive credentials through environment variables; credentials are not embedded in or stored with the artifact.